# Roton-like Reconstruction of the Emergent Photon in Quantum Spin Ice

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Quantum spin ice hosts an emergent photon at long wavelength, but its fate at lattice momenta is governed by a strongly interacting compact spin-$\frac{1}{2}$ gauge theory. We show that the pure pyrochlore ring-exchange model develops an order-one reconstruction of the finite-momentum $S^z$ spectrum. Operator-resolved calculations on 48-, 64-, 72-, and 96-site clusters reveal near the zone-boundary $L$ point a pair of dominant levels: a lower one at 0.90–1.07$K$ carrying 50–54% of the sublattice-summed spectral weight, and an upper one near 1.65–1.81$K$. The scale-free branch ratio $\omega_{\text{low}}/\omega_{\text{high}} = 0.55$–0.59 is stable across all four sizes. An infinitesimal Rokhsar–Kivelson deformation shows that the reconstructed and unreconstructed branches have opposite microscopic responses to ring flippability. Remarkably, the momentum dependence of the reconstruction is anticipated from the ground state alone: the Feynman ratio $f_1(\boldsymbol{q})/S(\boldsymbol{q})$ distinguishes reconstructing from nonreconstructing momentum classes and identifies a nearly degenerate soft mode away from $L$. Sign-free Green's-function Monte Carlo on complete momentum grids up to 2048 sites shows that the corresponding spectral centroid retains an $L$-centered minimum within a shallow zone-boundary valley. The resulting phenomenology is a lattice analogue of roton formation in a strongly interacting emergent gauge field.

Quantum spin ice provides a microscopic realization of an emergent compact $U(1)$ gauge field. At long wavelength its gapless excitation is governed by Maxwell electrodynamics, giving the familiar emergent photon of the Coulomb phase [1–4]. The microscopic theory, however, is not a weakly interacting Maxwell field. In the spin-$\frac{1}{2}$ ice manifold the electric field is saturated, $E = S^z = \pm\frac{1}{2}$, and the compact ring dynamics provide the entire nonconstant Hamiltonian. The lattice-scale continuation of the photon is therefore a genuinely interacting many-body problem.

This distinction is especially important experimentally. Large momentum does not by itself imply leaving the low-energy ice manifold: the effective ring Hamiltonian is a *lattice* Hamiltonian and therefore contains excitations throughout its Brillouin zone. What becomes nonuniversal at short wavelength is not the well-posedness of the problem but its sensitivity to microscopic terms beyond the minimal ring model. We therefore ask a sharply defined question: *what becomes of the emergent photon at lattice momentum within the minimal charge-free quantum-spin-ice Hamiltonian?*

The answer is unexpectedly dramatic. One of the two dominant collective branches undergoes an order-one reconstruction near the zone boundary. Approximately half of the spectral weight transfers into a mode near $1K$ in energy, while the upper branch remains almost a factor of two higher. The effect persists from 48 to 96 spins. Moreover, the reconstructed state has a distinct microscopic response to ring flippability, and the momentum sectors in which it appears can be identified from ground-state quantities alone.

Previous results already point toward strong finite-momentum photon interactions. The emergent fine-structure constant is large [5]; semiclassical approaches predict substantial anharmonic renormalization and interaction-induced splitting of the transverse polarizations [6–8]; and quantum Monte Carlo with analytic continuation finds broad zone-boundary response beyond Gaussian electrodynamics [9]. What has not been resolved is how the microscopic $S^z$ spectral weight is distributed among individual eigenstates. Our operator-resolved calculation determines that distribution directly.

We study the standard ring-exchange Hamiltonian in the spin-ice manifold,

$$H = -K \sum_p \left( W_p + W_p^\dagger \right), \qquad W_p = S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^-, \tag{1}$$

where $p$ runs over elementary contractible pyrochlore hexagons. In gauge language $W_p = e^{iB_p}$. Because

$$E_\ell = S_\ell^z = \pm\frac{1}{2}, \tag{2}$$

the conventional electric term $\sum_\ell E_\ell^2$ is a constant inside the constrained manifold. The photon stiffness is therefore generated dynamically by the ring Hamiltonian itself.

Our observable is the zero-temperature sublattice-summed pseudospin response

$$S^{zz}(\boldsymbol{q},\omega) = \frac{1}{N} \sum_{\mu,n} \left| \langle n | S_\mu^z(\boldsymbol{q}) | 0 \rangle \right|^2 \delta(\omega - \omega_n), \tag{3}$$

where

$$S_\mu^z(\boldsymbol{q}) = \sum_{i\in\mu} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i} S_i^z. \tag{4}$$

All spectral fractions quoted below are normalized within Eq. (3) at fixed momentum.

Equation (3) is the theoretical pseudospin response, not a complete material-specific neutron cross section. A physical neutron measurement additionally contains coherent sublattice interference, local easy-axis rotations, the magnetic $g$ tensor, form factors, and the neutron polarization projector. We return to this distinction below and give the full discussion in the Supplemental Material.

We evaluate Eq. (3) by complete diagonalization on 48 sites and Lanczos continued fractions on 64-, 72-, and 96-site clusters, reaching a constrained sector of dimension 146 758 352 at 96 sites. Cluster construction, winding-sector identification, momentum certification, and numerical convergence are detailed in the Supplemental Material.

Figure 1 shows the central result. Near the zone center, the dominant response follows the expected long-wavelength photon. At $L$, the spectrum changes qualitatively. The dominant low-energy levels occur at

$$\frac{\omega_{\mathrm{low}}}{K} = 1.024,\ 0.954,\ 1.068,\ 0.904 \tag{5}$$

for 48, 64, 72, and 96 sites, respectively. Their shares of the total $S^z$ weight at the same momentum are

$$50.8\%,\ 50.0\%,\ 54.0\%,\ 54.4\%. \tag{6}$$

A second dominant level remains at

$$\frac{\omega_{\mathrm{high}}}{K} = 1.730,\ 1.652,\ 1.814,\ 1.647. \tag{7}$$

Throughout this paper we call these two levels simply the *lower branch* and the *upper branch*. The terminology has a precise operational meaning and no more: at each momentum the $S^z$ probe lives in a two-dimensional transverse subspace of the emergent gauge field, and the two dominant levels couple through nearly orthogonal directions of that subspace (Supplemental Material), so they are the two independent collective features the probe can create there. No adiabatic connection to the free photon is implied by either name.

The most robust finite-size measure is the scale-free branch ratio,

$$\frac{\omega_{\mathrm{low}}}{\omega_{\mathrm{high}}} = 0.592,\ 0.578,\ 0.589,\ 0.549. \tag{8}$$

Both Eq. (6) and Eq. (8) remain nearly unchanged across three cluster shapes and a factor of roughly $1.6 \times 10^4$ in constrained Hilbert-space dimension. Neither requires comparison with a Gaussian fit.

The reconstruction is also not confined to the single point $L$. On the 72-site cluster,

$$\boldsymbol{q}_v = \left(\frac{1}{6}, \frac{1}{2}, \frac{1}{2}\right) \tag{9}$$

supports a dominant level at $1.075K$ carrying 55% of the $S^z$ weight, essentially degenerate with the $L$-point level at $1.068K$ carrying 54%. The finite-size spectra therefore indicate a soft zone-boundary region rather than an isolated anomalous momentum.

Fixing the single Gaussian energy scale from the long-wavelength spectrum places the Gaussian $L$ photon near $2.03K$. The reconstructed lower branch therefore lies at approximately one-half of the corresponding Gaussian energy. We use this comparison only to quantify the order-one magnitude of the effect; the finite-size stability itself follows directly from Eqs. (6) and (8).

The reconstructed branch is also microscopically distinct from the upper branch. To expose this difference without using the Gaussian benchmark, we employ the standard Rokhsar–Kivelson deformation

$$H(V) = H + V \sum_p n_p^{\mathrm{flip}}, \tag{10}$$

where $n_p^{\mathrm{flip}}$ projects onto a flippable alternating hexagon. We use $V$ only as an infinitesimal probe of the physical ring point $V = 0$. Hellmann–Feynman gives

$$\left.\frac{d\omega_n}{dV}\right|_{V=0} = \left\langle n \left| \sum_p n_p^{\mathrm{flip}} \right| n \right\rangle - \left\langle 0 \left| \sum_p n_p^{\mathrm{flip}} \right| 0 \right\rangle. \tag{11}$$

The slope is therefore an exact state-resolved census of the change in flippability carried by the excitation.

At the 48-site $L$ point, the reconstructed $1.024K$ level has

$$\frac{d\omega_{\mathrm{low}}}{dV} = +0.339, \tag{12}$$

whereas the upper branch at $1.730K$ responds with the opposite sign. More importantly, the same contrast appears in the anisotropic clusters that provide an internal control. On 64 sites, the reconstructing and nonreconstructing $L$ classes have censuses

$$+\,0.338 \qquad \text{and} \qquad -\,0.166, \tag{13}$$

respectively. On 96 sites the reconstructing class has

$$+\,0.259, \tag{14}$$

while the dominant levels of the nonreconstructing class have

$$-\,0.02, \qquad -0.09. \tag{15}$$

Thus momentum classes that are identical in $|\boldsymbol{q}|$ and in the Gaussian theory show opposite microscopic responses precisely according to whether their spectra reconstruct.

The positive census is a fingerprint rather than a definition of the mode. Its magnitude is less size-stable than the spectral weight or branch ratio, and positive values occur for other finite-size eigenstates — including the upper branch itself at 72 and 96 sites, as Fig. 3 shows. The exhaustive 48-site census and its exact trace identity are given in the Supplemental Material. The within-cluster sign contrast, however, provides an independent diagnostic that requires neither a fitted energy scale nor a comparison between different cluster sizes.

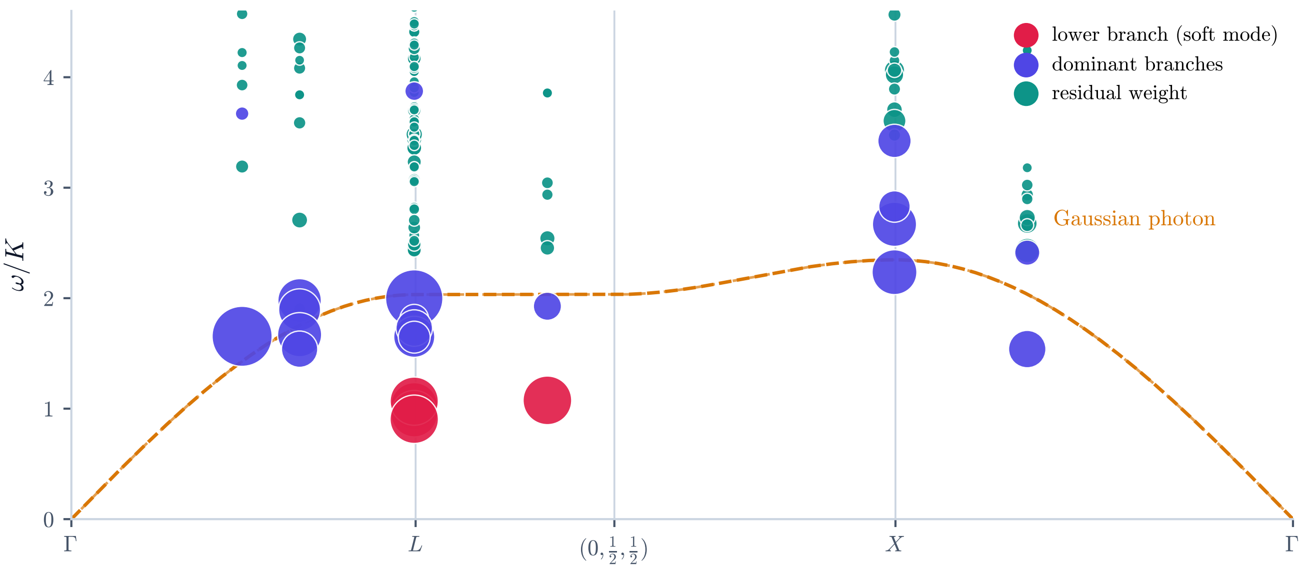


FIG. 1. **Nonperturbative finite-momentum reconstruction.** Operator-resolved $S^z$ spectrum along $\Gamma \to L \to (0, \frac{1}{2}, \frac{1}{2}) \to X \to \Gamma$, combining the diagonalized clusters. Marker area denotes the fraction of the sublattice-summed $S^z$ weight carried by each level at its momentum. The dashed curve is the Gaussian photon with a single fitted overall energy scale. Red marks the lower branch where it reconstructs; indigo marks the dominant branches elsewhere and the upper branch; teal is residual weight. The long-wavelength response is photon-like, whereas near $L$ the lower branch reconstructs into a dominant low-energy level. A similar reconstruction occurs at $(\frac{1}{6}, \frac{1}{2}, \frac{1}{2})$ on the 72-site cluster.

We now ask why the reconstruction appears where it does. The first frequency moment of Eq. (3) can be evaluated exactly,

$$f_1(\boldsymbol{q}) = \frac{K}{2N} \sum_p \langle W_p + W_p^\dagger \rangle \sum_\mu \left| g_p^\mu(\boldsymbol{q}) \right|^2, \tag{16}$$

where $g_p^\mu(\boldsymbol{q})$ is the alternating lattice curl of the sublattice-resolved electric-field wave around plaquette $p$. Together with

$$S(\boldsymbol{q}) = \int d\omega\, S^{zz}(\boldsymbol{q}, \omega), \tag{17}$$

this gives the Feynman single-mode ratio

$$\omega_{\mathrm{SMA}}(\boldsymbol{q}) = \frac{f_1(\boldsymbol{q})}{S(\boldsymbol{q})}. \tag{18}$$

Equation (18) is exactly the spectral centroid of the state created by $S^z(\boldsymbol{q})$. A small value therefore does not by itself prove that the spectrum contains a single sharp pole. Its importance here is different: both $f_1$ and $S$ are properties of the ground state. The ratio can consequently be evaluated before calculating any excited state and tested as a forward diagnostic of where strong spectral reconstruction occurs.

The anisotropic clusters provide particularly sharp tests. On 64 sites, the $L$ star splits into two inequivalent momentum classes having the same $|\boldsymbol{q}|$ and the same Gaussian energy. Their interacting ground-state structure factors are

$$S = 0.369 \qquad \text{and} \qquad S = 0.227, \tag{19}$$

giving substantially different values of $f_1/S$. The lower-ratio class subsequently develops the $0.954K$ level carrying 50.0% of the weight. The other remains dominated by a $1.998K$ level carrying 77%. The reconstruction therefore follows the interacting ground-state diagnostic rather than the label $L$ itself.

The same test repeats on 96 sites. Its two inequivalent $L$ classes have

$$S = 0.355 \qquad \text{and} \qquad S = 0.241. \tag{20}$$

The lower-ratio class develops the $0.904K$ level with 54.4% of the weight, whereas the other retains its dominant response near 1.65–$1.73K$. Subject to the 96-site sector and Krylov checks noted above, this is an independent repeat of the 64-site discrimination.

A different test occurs on 72 sites. The ground-state values of $f_1/S$ at $L$ and at $(\frac{1}{6}, \frac{1}{2}, \frac{1}{2})$ differ by only 0.9%. Their dominant low-energy levels subsequently appear within 0.7%:

$$\omega_L = 1.068K, \qquad \omega_{\boldsymbol{q}_v} = 1.075K. \tag{21}$$

Their corresponding spectral-weight fractions are 54% and 55%. The ratio does not predict either absolute pole

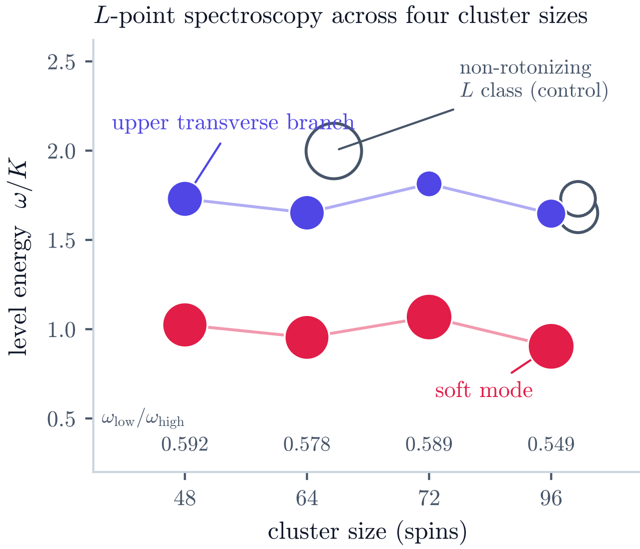


FIG. 2. **Finite-size stability and internal controls.** Filled symbols show the lower and upper branches of the reconstructing $L$ class. Open symbols show the dominant levels of the inequivalent nonreconstructing $L$ class on the anisotropic 64- and 96-site clusters. The numbers denote $\omega_{\mathrm{low}}/\omega_{\mathrm{high}}$. The lower branch of the reconstructing class retains approximately half of the sublattice-summed $S^z$ weight at every size.

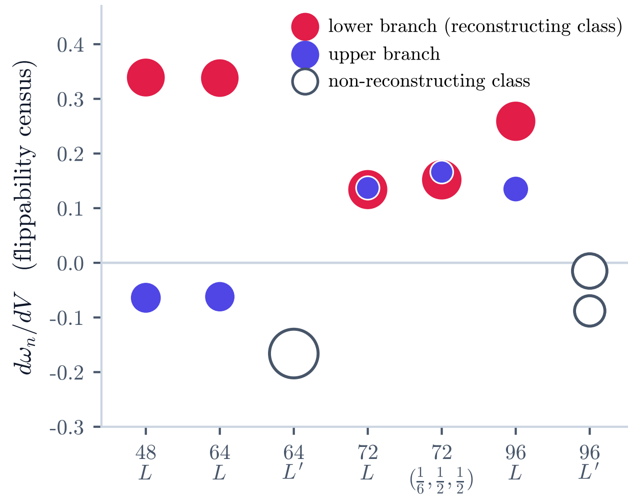


FIG. 3. **Microscopic fingerprint of the reconstruction.** The RK derivative $d\omega_n/dV$ [Eq. (11)] measures the excess flippability carried by each excitation. The robust statement is the *within-cluster class contrast*: on the anisotropic 64- and 96-site clusters, the lower branch of the reconstructing $L$ class carries a positive census while the dominant levels of the nonreconstructing class (open symbols), at the same $|\boldsymbol{q}|$, carry negative ones. The sign of the *upper* branch is not itself a diagnostic: it is negative at 48 and 64 sites but positive at 72 and 96, one instance of the census magnitudes' limited size stability discussed in the text. The full census distribution and deformation trajectories are given in the Supplemental Material.

energy: the centroid also contains the high-energy spectral weight. It does, however, identify two distinct momentum sectors with almost identical ground-state centroids, both of which subsequently show strongly reconstructed spectra.

The small ratio near $L$ has two ingredients. The first is geometric. For $\boldsymbol{q} \parallel [111]$, one family of pyrochlore hexagons lies in the wavefronts of the electric-field oscillation. The alternating lattice curl through those plaquettes vanishes exactly, reducing the oscillator-strength cost in Eq. (16). At $L$ the family-resolved geometric weights are

$$(96, 96, 96, 0), \tag{22}$$

compared with

$$(96, 96, 96, 96) \tag{23}$$

at $X$.

Geometry alone is not sufficient. The same exact cancellation occurs at shorter wavelength along [111], where no comparable reconstruction appears. The second ingredient is the enhanced interacting structure factor. On the 48-site cluster,

$$S(L) = 0.370, \tag{24}$$

whereas the equal-amplitude superposition over the same constrained configurations gives

$$S_{\mathrm{EA}}(L) = 0.297. \tag{25}$$

Classical loop Monte Carlo on a 2048-site ice manifold likewise gives an essentially flat structure factor among the principal zone-boundary momenta. The $L$ enhancement is therefore generated predominantly by the interacting quantum amplitudes rather than by the ice constraint alone.

The RK trajectory provides a complementary check. Biasing toward more flippable configurations, $V < 0$, increases $S(L)$ and softens the lower branch from approximately $1.02K$ to $0.73K$, while $f_1(L)$ changes by less than one percent. We interpret this as a correlated evolution of the ground-state structure factor and the soft spectrum, not as a temporal or causal ordering. The complete deformation scan is shown in the Supplemental Material.

Because Eq. (18) contains ground-state quantities only, its momentum dependence can be evaluated on systems far beyond the reach of operator-resolved diagonalization. The ring Hamiltonian is stoquastic in the ice basis, allowing sign-free Green's-function Monte Carlo with forward walking. The method is benchmarked against exact ground-state quantities on 48 sites and against the 72-site ground-state energy; convergence in walker number and forward-walking time is given in the Supplemental Material.

We compute $S(\boldsymbol{q})$ on the complete allowed momentum grids of $6^3$ and $8^3$ cubic supercells containing 864 and 2048 spins. At the principal zone-boundary points,

$$\begin{array}{c|cc} & 864 & 2048 \\ \hline L & 2.142(10) & 2.179(19) \\ X & 2.660(10) & 2.713(21) \\ W & - & 2.781(10) \\ K & - & 2.718(20) \\ U & - & 2.736(11) \end{array} \tag{26}$$

in units of $K$. The $L$ value changes by only 1.7% between 864 and 2048 sites and remains 20–25% below the other principal zone-boundary momenta.

The large-system result in Fig. 4 is a statement about the *spectral centroid*, not the thermodynamic survival of an individual sharp pole. Along the radial [111] direction, $f_1/S$ rises from $\Gamma$ to $L$. The centroid therefore does not form a helium-like radial minimum. Within the zone-boundary manifold, however, it develops an $L$-centered minimum and a shallow valley along $(t, \frac{1}{2}, \frac{1}{2})$. The valley is distinguished at matched $|\boldsymbol{q}|$, not merely by a low value: interior momenta at intermediate $|\boldsymbol{q}|$, such as $(0, 0, \frac{3}{4})$, also lie below the boundary plateau, but only because the centroid is still rising along their lines; at the same $|\boldsymbol{q}| = 0.75$ the valley sits 9–10% below $(0, 0, \frac{3}{4})$, far outside the statistical errors, and is nearly flat along its own line. The 72-site spectrum at $(\frac{1}{6}, \frac{1}{2}, \frac{1}{2})$ provides a direct finite-size dynamical example that this low-centroid valley can host strongly reconstructed spectral weight away from $L$.

The resulting phenomenology is naturally roton-like. In Feynman's construction, a collective excitation becomes inexpensive where the ratio of oscillator strength to static correlation is small. Here the same organizing principle identifies a weight-dominant finite-momentum excitation of a compact emergent gauge field. The designation attaches to the lower branch only, and for measured reasons rather than by definition. The upper branch is not anomalously soft: at 1.65–1.81$K$ it sits at 81–89% of the Gaussian energy, the size of renormalization an ordinarily interacting photon exhibits, whereas the lower branch sits near one half. It is not weight-dominant: it retains 18–32% of the response against 50–54%. And it is not the state the Feynman construction produces: the single-mode state built from $S^z(\boldsymbol{q}_L)|0\rangle$ overlaps the lower-branch multiplet with squared amplitude 0.88, 0.87, and 0.86 on the 48-, 64-, and 72-site clusters, coupling to the upper branch only through the orthogonal transverse channel. A shared census sign at some sizes (Fig. 3) alters none of this, the census being a fingerprint rather than a definition. The lattice geometry changes the detailed shape of the phenomenon: rather than a continuum radial dip identical to that of liquid helium, the finite-size branch descends into the zone boundary while the large-system centroid develops a minimum within the boundary manifold. We use "roton-like" in this operational sense and do not assume that the finite-size pole remains infinitely sharp in the thermodynamic limit.

The interpretation also does not require a strict adiabatic "photon ancestry" for the low level. The $S^z$ probe occupies a two-dimensional transverse gauge-field subspace, and the two dominant levels couple through nearly orthogonal transverse directions. This is the operational content of the lower- and upper-branch terminology used throughout. The basis-invariant polarization construction is given in the Supplemental Material.

Additional state-resolved diagnostics support the collective character of the reconstructed branch. A variational electric-field wave relaxes rapidly toward the exact 48-site state, and an exact basis-invariant plaquette decomposition shows that the reconstructed mode recovers 28–32% of its bare single-mode energy on the 48-, 64-, and 72-site clusters. The recovery is strongly selective among hexagon families but is not concentrated in the geometrically dark family. Parity, weak-photon kinematics, off-pole spectral weight, and the full family decomposition are presented in the Supplemental Material.

For experiment, the robust conclusion is spectral redistribution rather than a universal intensity ratio. The pure ring model shows that an emergent photon that appears as a single long-wavelength mode can reorganize strongly at lattice momenta without leaving the charge-free ice manifold. Near the zone boundary, a lower feature acquires a large fraction of the theoretical $S^z$ response while the upper branch remains at higher energy, and additional weight appears outside the two dominant branches.

A material-specific neutron cross section may reweight these features substantially. The experimental intensity is a coherent projection of the sublattice-resolved amplitudes and depends on the local easy axes, $g$ tensor, magnetic form factor, and neutron polarization. Consequently the theoretical 50–54% weight fraction should not be read as a universal neutron intensity fraction. Likewise, longer ring exchanges, virtual spinons, and other material-specific interactions can modify the zone-boundary spectrum. The experimentally robust message of the minimal model is instead that strong short-wavelength reconstruction is possible already within the compact gauge sector itself.

In summary, the emergent photon of the pure quantum-spin-ice ring model is nonperturbatively reconstructed at the lattice scale. Across four finite clusters, the $S^z$ response near the zone boundary transfers approximately half of its spectral weight into a lower branch with a stable scale-free ratio to the upper one,

$$\omega_{\text{low}}/\omega_{\text{high}} = 0.55\text{–}0.59. \tag{27}$$

The reconstructed state has an independent microscopic fingerprint in its response to ring flippability. The ground-state Feynman ratio identifies where the reconstruction occurs, including inequivalent $L$ classes and a

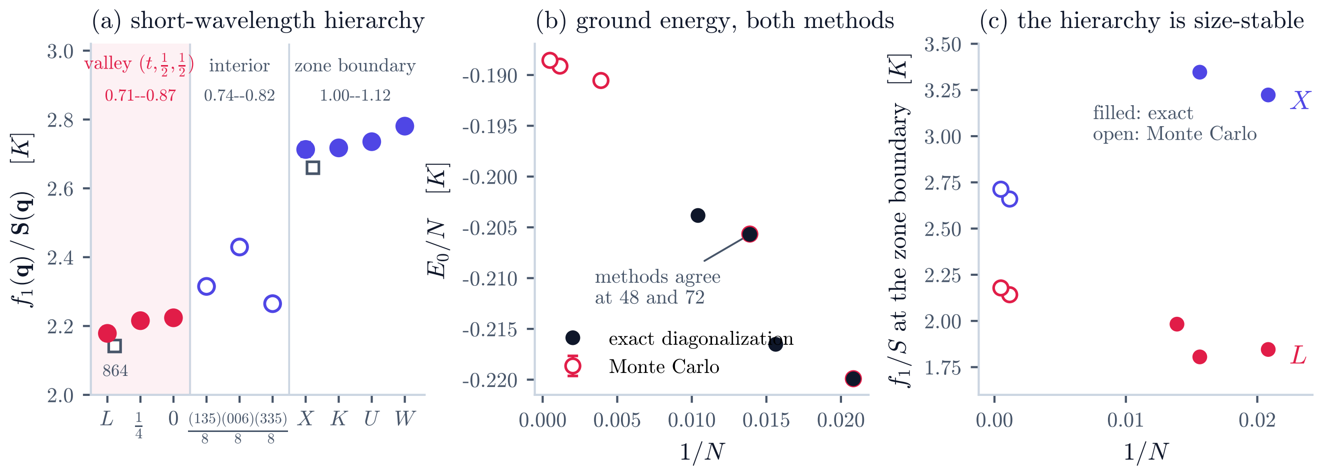


FIG. 4. **Large-system ground-state landscape.** (a) Feynman ratio $f_1(\boldsymbol{q})/S(\boldsymbol{q})$ on the 2048-site supercell, grouped by geometric role, with each group's $|\boldsymbol{q}|$ range (cubic reciprocal units) indicated. Red: the $(t, \frac{1}{2}, \frac{1}{2})$ valley, nearly flat along its line. Open circles: *interior* stars at comparable $|\boldsymbol{q}|$ — these sit below the boundary plateau only because the branch is still rising with $|\boldsymbol{q}|$; at matched $|\boldsymbol{q}| = 0.75$ the valley lies 9–10% (many standard deviations) below the interior star $(0, 0, \frac{3}{4})$, which is mid-climb along $\Delta$, not a second valley. Filled indigo: the true zone-boundary momenta $X$, $K$, $U$, $W$, 20–25% above $L$. Open squares: 864-site values at the momenta the two grids share. The complete allowed momentum grid is shown in the Supplemental Material. (b) Ground-state-energy benchmarks comparing exact diagonalization and GFMC. (c) Size evolution of the ratio at $L$ and $X$ across both methods. GFMC constrains the spectral centroid, whereas the finite-cluster diagonalizations establish the dominant low-energy pole.

soft valley away from $L$, while sign-free Monte Carlo shows that the associated centroid landscape remains robust on complete momentum grids up to 2048 spins. These results establish a lattice-scale regime in which the dynamics of a compact spin-$\frac{1}{2}$ gauge field are qualitatively richer than those of a weakly renormalized Maxwell photon.

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